

INNER PRODUCT

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. A vector space $\mathcal{V}$ over $\mathbb{F}$ is said to be an inner-product space if it is equipped with a function called inner product

$$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{F} \quad (\mathbb{F} = \mathbb{R} \text{ or } \mathbb{C}).$$

which satisfies the following properties.

$$(i) \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle, \quad \forall u, v, w \in \mathcal{V}.$$

$$(ii) \langle \alpha u, v \rangle = \alpha \langle u, v \rangle, \quad \forall u, v \in \mathcal{V}, \alpha \in \mathbb{F}.$$

$\uparrow$
multiplication of real or complex numbers

$$(iii) \langle u, v \rangle = \overline{\langle v, u \rangle} \quad (\bar{\alpha} \text{ represents complex conjugate of } \alpha)$$

$\forall u, v \in \mathcal{V}$

$$(iv) \langle u, u \rangle \geq 0, \quad \forall u \in \mathcal{U}. \text{ Moreover, } \langle u, u \rangle = 0 \text{ if and only if } u = 0$$

Example:

$$(i) \mathcal{V} = \mathbb{C}^n$$

Define

$$\langle u, v \rangle = v^H u \quad \left(= \sum_{i=1}^n v_i^* u_i \right)$$

Verify it is an inner product w.r.t. $\mathbb{C}$

$$(ii) \mathcal{V} = \mathbb{R}^n$$

Define

$$\langle u, v \rangle = v^T u \quad \left(= \sum_{i=1}^n u_i v_i \right)$$

Verify it is an inner product w.r.t. $\mathbb{R}$

$$(iii) \mathcal{V} = \mathbb{C}^{m \times n}$$

Define

$$\langle U, V \rangle = \text{Trace} \left(\overbrace{V^H U}^{n \times n} \right)$$

Can be shown that this defines an inner product on $\mathbb{C}^{m \times n}$

$$U, V \in \mathbb{C}^{m \times n}$$

Here $V^H \in \mathbb{C}^{n \times m}$, defined as

$$[V^H]_{ij} = \overline{V_{ji}} \quad \left(\text{transposed \& conjugated} \right)$$

Given a square matrix A ($n \times n$)

$$\text{Trace}(A) = \sum_{i=1}^n A_{ii} \quad \left(\text{sum of diagonal elements of } A \right)$$

Norm Induced By Inner Product.

Let V be equipped with inner product $\langle \cdot, \cdot \rangle$.

Define a function

$$\| \cdot \|_{ip} : V \rightarrow \mathbb{R}^+ \text{ as}$$

$$\|v\|_{ip} = \sqrt{\langle v, v \rangle}, \quad v \in V$$

$\| \cdot \|_{ip}$ is called the "norm induced by the inner product".

[Can be shown why it is a "norm".
 $\Rightarrow$ See HW]

CAUCHY-SCHWARZ INEQUALITY

Let $\mathcal{V}$ be an inner product space.

Then, for any $u, v \in \mathcal{V}$, we have

$$|\langle u, v \rangle| \leq \|u\|_{ip} \|v\|_{ip}$$

where $\|u\|_{ip} = \sqrt{\langle u, u \rangle}$.

Notion of Angle:

$$\mathcal{V} = \mathbb{R}^n$$

$$\langle u, v \rangle = v^T u = v^T v, \quad \|v\|_{ip} = \left(\sum_{i=1}^n |v_i|^2 \right)^{1/2}$$

$$-1 \leq \frac{\langle u, v \rangle}{\|u\|_2 \|v\|_2} \leq 1$$

$\triangleq \cos(\theta)$

θ is called the "angle" between vectors u, v .

Inner product (via CS inequality)

leads to a notion of "angle"

between vectors u, v .

$$\begin{aligned} \text{(i) } u=v \Rightarrow \cos \theta &= \frac{\langle u, u \rangle}{\|u\|_{ip} \|v\|_{ip}} \\ &= \frac{\|u\|_{ip}^2}{\|u\|_{ip}^2} \\ &= 1 \end{aligned}$$

$$\Rightarrow \theta = 0$$

$$\begin{aligned} \text{(ii) } u=-v, \cos \theta &= -\frac{\langle u, u \rangle}{\|u\|_{ip} \|u\|_{ip}} \\ &= -1 \end{aligned}$$

$$\Rightarrow \theta = 180^\circ$$

ORTHOGONALITY

A set of ^{non-zero} vectors $v_1, v_2, \dots, v_k$ are orthogonal if

$$\langle v_i, v_j \rangle = 0, \text{ whenever } i \neq j$$

[Any subset of $\{v_1, \dots, v_k\}$ will also be orthogonal]

Example:

(i) $V = \mathbb{C}^n$

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

are

orthogonal.

(ii) $v_1 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix},$

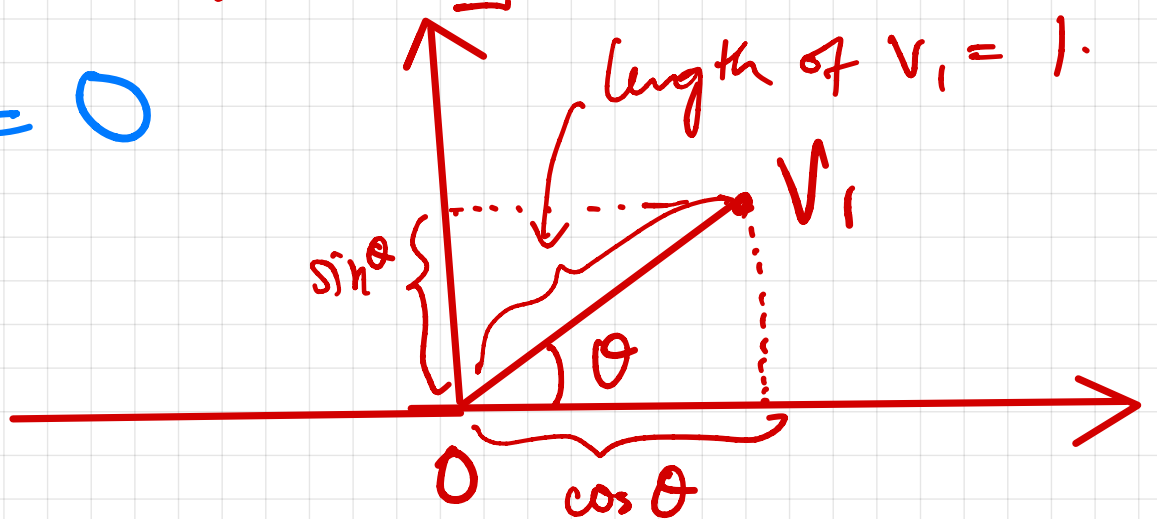
$v_2 = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$

Place v_2 in the figure (H.W.)
($V = \mathbb{R}^2$)

Can show

$$v_1^T v_2 = 0$$

$$\|v_1\|_2 = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$



(iii) Let $A \in \mathbb{C}^{m \times n}$, where $a_1, a_2, \dots, a_n$ are orthogonal.

Define $A^H \in \mathbb{C}^{n \times m}$ as the Hermitian of A (conjugate transpose)

$$\text{ie } [A^H]_{ij} = \overline{A_{ji}}, \quad \begin{matrix} i=1,2,\dots,n \\ j=1,2,\dots,m \end{matrix}$$

Then, we have

$$\underbrace{A^H}_{n \times m} \underbrace{A}_{m \times n} = \begin{bmatrix} a_1^H \\ a_2^H \\ \vdots \\ a_n^H \end{bmatrix} \begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix}$$

$$= \begin{bmatrix} a_1^H a_1 & a_1^H a_2 & a_1^H a_3 & \dots & a_1^H a_n \\ a_2^H a_1 & a_2^H a_2 & a_2^H a_3 & \dots & a_2^H a_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_n^H a_1 & a_n^H a_2 & a_n^H a_3 & \dots & a_n^H a_n \end{bmatrix}$$

$$= \begin{bmatrix} \|a_1\|_2^2 & & & & \\ & \|a_2\|_2^2 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & \|a_n\|_2^2 \end{bmatrix}$$

If $\|a_i\|_2 = 1, \quad i = 1, 2, \dots, n.$

then $A^H A = I_{n \times n}$

(However, we may
not have
 $AA^H = I$)

Such a matrix $A \in \mathbb{C}^{m \times n}$ is said to
have orthonormal columns

Unitary matrix:

A matrix $A \in \mathbb{C}^{n \times n}$ is unitary if it has orthogonal columns with unit norm. A is square.

ie $A = [a_1 \ a_2 \ \dots \ a_n]$, $a_i \in \mathbb{C}^n$

$$\langle a_i, a_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

The above can be written in a compact way as

$$A^H A = I$$

$\Rightarrow A^H$ is "the" inverse of $A \Rightarrow A A^H = A^H A = I$

Equivalent Definition

A matrix $A \in \mathbb{C}^{n \times n}$ is unitary iff

$$A^H A = A A^H = I_{n \times n}.$$

IMPLICATIONS OF ORTHOGONALITY.

① If $v_1, v_2, \dots, v_k$ are orthogonal,
then $v_1, v_2, \dots, v_k$ are also linearly
independent.

Proof:

Suppose

$$\sum_{i=1}^k \alpha_i v_i = 0, \quad \alpha_i \in \mathbb{F}$$

verify that $\langle 0, v \rangle = 0$

$$\Rightarrow \left\langle \sum_{i=1}^k \alpha_i v_i, v_1 \right\rangle = \langle 0, v_1 \rangle = 0$$

$$\Rightarrow \sum_{i=1}^k \alpha_i \underbrace{\langle v_i, v_1 \rangle}_{=0, i \neq 1} = 0$$

$$\Rightarrow \alpha_1 \|v_1\|_{ip}^2 = 0$$

Since $v_1 \neq 0 \Rightarrow \alpha_1 = 0$. Similarly show
that $\alpha_i = 0, i = 1, 2, \dots, k$.

ORTHOGONAL BASIS

Suppose S is a subspace with $\dim(S) = n$

Suppose $\{v_1, \dots, v_n\}$ is an orthogonal set of vectors in S , i.e. $v_i \in S, i=1, \dots, n$ and

$$\langle v_i, v_j \rangle = 0, \quad i \neq j$$

Then $\{v_1, \dots, v_n\}$ is a linearly independent set in S and therefore must be a basis of S ($\dim(S) = n$).

$\{v_1, \dots, v_n\}$ is called an orthogonal basis of S .

Example: DFT Basis

Consider $w_1, w_2, \dots, w_N \in \mathbb{C}^N$, defined as

$$w_i = \frac{1}{\sqrt{N}} \begin{bmatrix} e^{j \frac{2\pi(i-1)}{N}} \\ e^{j \frac{2\pi \cdot (2(i-1))}{N}} \\ \vdots \\ e^{j \frac{2\pi(m \cdot (i-1))}{N}} \\ \vdots \\ e^{j \frac{2\pi(N-1)(i-1)}{N}} \end{bmatrix}, \quad i = 1, 2, \dots, N$$

① Show that $\{w_1, w_2, \dots, w_N\}$ is an orthonormal set of vectors.

This will imply that $\{w_1, \dots, w_{N-1}\}$ is an orthonormal BASIS of $\mathbb{C}^N$.

② Let $v \in \mathbb{C}^N$

$$v = \sum_{i=1}^N \alpha_i w_i$$

Convince yourself that

$$\alpha_i = \frac{\langle v, w_i \rangle}{\|w_i\|_2^2}$$

in $\mathbb{C}^N$,
 $\left(\begin{aligned} \langle v, w_i \rangle \\ = w_i^H v \end{aligned} \right)$

And convince yourselves that

α_i = i^{th} DFT coefficient of the signal v .

→ This will represent the Discrete Fourier Transform of v .

Inverse DFT:

$$\alpha_i = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} v_n e^{-j \frac{2\pi n(i-1)}{N}}$$

$$v = \sum_{i=1}^N \alpha_i w_i = \sum_{i=1}^N \frac{\langle v, w_i \rangle}{\|w_i\|_2^2} w_i$$

Same idea Applies to any other orthogonal Basis of $\mathbb{C}^N$.